

CHEMISTRY

TARGET : JEE- Advanced

**CAPS-23
SOLID STATE****SINGLE CHOICE QUESTIONS**

- Sodium ($\text{Na} = 23$) crystallizes in bcc arrangement with the interfacial separation between the atoms at the edge 53.6 pm. The density of sodium crystal is
(A) 2.07 g/cc (B) 2.46 g/cc (C) 1.19 g/cc (D) none of these
- An element X (atomic weight = 24 gm/mol) forms a face centered cubic lattice. If the edge length of the lattice is 4×10^{-8} cm and the observed density is $2.40 \times 10^3 \text{ kg/m}^3$, then the percentage occupancy of lattice points by element X is (Use $N_A = 6 \times 10^{23}$)
(A) 96 (B) 98 (C) 99.9 (D) none of these
- Metallic Gold crystallizes in fcc lattice and the length of cubic unit cell is 407 pm.
(Given: Atomic mass of Gold = 197, $N_A = 6 \times 10^{23}$)
The density if it has 0.2% Schottky defect is (in gm/cm^3):
(A) 4.86 (B) 9.72 (C) 19.48 (D) 18.44
- Select the correct statement(s).
(A) The co-ordination number of each type of ion in a CsCl crystal is twelve
(B) A metal that crystallizes in a bcc structure has a co-ordination number of twelve
(C) A unit cell of an ionic crystal shares some of its ions with other unit cells
(D) The length of the unit cell in NaCl is 552 pm (given that $r_{\text{Na}^+} = 85 \text{ pm}$ and $r_{\text{Cl}^-} = 181 \text{ pm}$)
- A big RED spherical balloon (radius = $6a$) is filled up with gas. On this balloon six small GREEN spherical balloons (radius = a) are stuck on the surface in a specific manner. As RED balloon is slowly deflated, a point comes when all these six GREEN balloons touch and green balloons arrange themselves in a 3-D closed packing arrangement. At that stage, the radius of the RED balloon would have reduced by approximately
(A) 14.5 times (B) 1.414 times (C) 6.0 times (D) 2.42 times
- 100 cc of a piece of impure rock salt density 4 g/cc is dissolved in water and treated with excess of silver nitrate so that all chloride is precipitated as AgCl. If the number of moles of AgCl precipitated is 6, the percentage purity of the sample is
(A) 43.9 (B) 87.75 (C) 90 (D) 55.5

MULTIPLE CHOICE QUESTIONS

7. Pick out the correct statement(s).
- (A) Due the presence of F-centers in crystals, the crystal attains paramagnetic character.
- (B) Frenkel defect is a dislocation defect.
- (C) In case of diamond that follows zinc blende structure, the number of carbon atoms per unit cell is four.
- (D) Due to Schottky defect in crystals, the electrical neutrality is not affected.
8. Select the correct statement(s).
- (A) Co-ordination no. of Cs^+ and Cl^- are 8, 8 in CsCl crystal
- (B) If radius ratio (r_c/r_a) < 0.225 then shape of compound must be linear
- (C) If radius ratio (r_c/r_a) lies between 0.414 to 0.732 then shape of ionic compound may be square planer (Ex. PtCl_4^{2-})
- (D) If radius ratio is less than 0.155 then shape of compound is linear
9. Select the correct statement(s).
- (A) CsCl changes to NaCl structure on heating
- (B) NaCl changes to CsCl structure on applying pressure
- (C) Co-ordination number decreases on applying pressure
- (D) Co-ordination number increases on heating
10. Amorphous solids
- (A) do not have sharp melting points.
- (B) are isotropic.
- (C) have same physical properties in all directions.
- (D) are supercooled liquids.
11. Which is/are correct statement about zinc blende structure?
- (A) The number of first neighbors of S^{2-} is 4.
- (B) The maximum distance between Zn^{2+} is $\frac{a\sqrt{3}}{2}$, where 'a' = edge length of unit cell.
- (C) If all tetrahedral voids occupied by Zn^{2+} then C.N. of S^{2-} is 8.
- (D) If all tetrahedral voids occupied by Zn^{2+} then C.N. change from 4 : 4 to 8 : 8.

Paragraph for (Ques. 12 to Q. 14)

In HCP as well as CCP, only 74% of the available space is occupied by spheres. The remaining space is vacant and constitute interstitial voids or spaces. There are two types of interstitial space, tetrahedral and octahedral voids, in three-dimensional close packing. In a close packing, the number of tetrahedral voids is double the number of spheres. Radius of the tetrahedral voids relative to the radius of the sphere is 0.225, that is, for tetrahedral voids $\frac{r_{\text{void}}}{r_{\text{sphere}}} = 0.225$. In a close packing, the number of octahedral voids is equal to the number of spheres. Radius of the octahedral void relative to the radius of the sphere is 0.414 that is, for octahedral void $\frac{r_{\text{void}}}{r_{\text{sphere}}} = 0.414$.

12. In a compound of XY_2O_4 , oxide ions are arranged in CCP and cations X are present in octahedral voids. Cations Y are equally distributed between octahedral and tetrahedral voids. The fraction of the octahedral voids occupied is
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$
13. In a solid, oxide ions are arranged in CCP, cations A occupy $\frac{1}{6}$ of the tetrahedral voids and cations B occupy $\frac{1}{3}$ of the octahedral voids. The formula of the compound is
 (A) ABO_3 (B) AB_2O_3 (C) A_2BO_3 (D) $A_2B_2O_3$
14. Mineral having the formula AB_2 crystallizes in cubic packed lattice, with A atoms occupying the lattice points. The coordination number of A atoms, that of B atoms and percentage of tetrahedral voids occupied by B atoms are respectively.
 (A) 8, 4, 100% (B) 2, 6, 75% (C) 3, 1, 25% (D) 6, 6, 50%

Paragraph for (Ques. 15 to Q. 17):

Diamond is a crystalline allotrope of carbon, which crystallizes in a 3-dimensional lattice. Each carbon forms four bonds with other carbon atoms. The unit cell of the diamond lattice can be thought of containing carbon atoms present at all CCP positions as well in alternate tetrahedral voids. Assuming contact between nearest atoms, answer the following:

15. Number of carbon atoms per unit cell of diamond structure is
 (A) 4 (B) 8 (C) 12 (D) data insufficient
16. What is the fraction of length covered along the body diagonal of the cubic unit cell by carbon atoms?
 (A) 0.50 (B) 0.34 (C) 0.75 (D) 0.25
17. How much fraction of area of one face is covered by atoms?
 (A) $\frac{3\pi}{32}$ (B) $\frac{6\pi}{32}$ (C) $\frac{9\pi}{32}$ (D) $\frac{3\pi}{64}$

NUMERICAL VALUE TYPE

18. In seven possible crystal system how many crystal systems have more than one Bravais lattice?
19. Ionic solid $\text{Na}^+ \text{A}^-$ crystalline in rock salt type structure. 2.592 gm of ionic solid salt NaA dissolved in water to make 2 litre solution. The pH of this solution is 8. If distance between cation and anion is 300 pm. Calculate density of ionic solid (in gm/cm^3).
(Given: $\text{pK}_w = 13$, $\text{pK}_a(\text{HA}) = 5$, $N_A = 6 \times 10^{23}$)
20. In a solid 'AB' having NaCl structure, A atoms are in FCC. If all the face centered atoms along one of the axes are removed, then what will be the sum of atoms in a unit cell of compound AB?
21. A strong current of trivalent gaseous boron passed through a germanium crystal decreases the density of the crystal due to part replacement of germanium by boron and due to interstitial vacancies created by missing Ge atoms. In one such experiment, one gram of Germanium is taken, and the boron atoms are found to be 150 ppm by weight when the density of the Ge crystal decreases by 4%. Calculate the percentage of missing vacancies due to germanium which are filled up by boron atoms.
Atomic wt. Ge = 72.6, B = 11
22. It is believed that non-stoichiometric compound $\text{Fe}_{0.93}\text{O}$ forms by doping of Fe^{3+} ions in FeO crystal by replacement of Fe^{2+} . Calculate the total no. of cationic vacancies if now all the Fe^{2+} ions are replaced by Si^{4+} ions in 0.1 mole of $\text{Fe}_{0.93}\text{O}$. (Give your answer in multiple of N_A)

MATRIX MATCH TYPE

23. Match the following.

Column-I

[Bravais Lattice (s)]

(A) Primitive, face centered, body centered, end centered

(B) Primitive, face centered, body centered

(C) Primitive, body centered

(D) Primitive only

(A) $A \rightarrow S$; $B \rightarrow P$; $C \rightarrow R$; $D \rightarrow Q$

(B) $A \rightarrow P$; $B \rightarrow Q$; $C \rightarrow R$; $D \rightarrow S$

(C) $A \rightarrow R$; $B \rightarrow P$; $C \rightarrow S$; $D \rightarrow Q$

(D) $A \rightarrow Q$; $B \rightarrow P$; $C \rightarrow S$; $D \rightarrow R$

Column-II

(Crystal System)

(P) Cubic

(Q) Orthorhombic

(R) Hexagonal

(S) Tetragonal

24. Match the following.

Column-I (Structure)

- (A) Rock salt
 (B) Zinc blende
 (C) Fluorite
 (D) Anti fluorite (Na_2O)
 (A) $A \rightarrow S$; $B \rightarrow P$; $C \rightarrow Q$; $D \rightarrow R$
 (C) $A \rightarrow R$; $B \rightarrow S$; $C \rightarrow Q$; $D \rightarrow P$

Column-II (Voids occupied)

- (P) 100% tetrahedral voids occupied by cation
 (Q) 100% tetrahedral voids occupied by anion
 (R) 100% octahedral voids occupied by cation
 (S) 50% tetrahedral voids occupied by cation
 (B) $A \rightarrow P$; $B \rightarrow R$; $C \rightarrow S$; $D \rightarrow Q$
 (D) $A \rightarrow Q$; $B \rightarrow P$; $C \rightarrow S$; $D \rightarrow R$

Answer Key CAPS-23

1. (C)	2. (A)	3. (C)	4. (C)	5. (A)
6. (B)	7. (ABD)	8. (ACD)	9. (AB)	10. (ABCD)
11. (AC)	12. (A)	13. (A)	14. (A)	15. (B)
16. (A)	17. (A)	18. (4.00)	19. (4.00)	20. (7.00)
21. (2.367%)	22. (0.0465 N_A)	23. (D)	24. (C)	